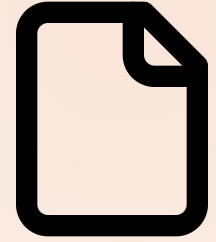


MODULE 08 · 24 MIN

Refactoring & Documentation at Scale



Refactor under written constraints. Document from the diff — never from the prompt.

Theory · Constrained refactor + two-pass docs (4 min)

Tell Claude what may NOT change: public API, file count, runtime behavior.

- **Two-pass workflow** — keep them separate:
 - **Pass 1:** refactor for readability only.
 - **Pass 2:** generate docs **from the diff**.
- **HANDOFF.md** — one-pager for the next engineer: what changed · why · risk · watch-outs.
- **ARCHITECTURE.md** — component shape, data flow, **one** diagram (ASCII is fine), known limits.

Combine the two passes and the docs describe your prompt, not the code.

Refactor inside the guardrails

REFACTOR UNDER WRITTEN CONSTRAINTS

≤ 60 lines / function

No new deps

Public API unchanged

BEFORE

```
def handle(x):  
  # 180 lines  
  # nested ifs  
  # 4 globals  
  ...  
  
cyclomatic: 24
```



AFTER

```
def handle(x):  
  v = validate(x)  
  d = build(v)  
  return send(d)  
  
cyclomatic: 4
```

constraints.md fixes what must NOT change — write it *before* you touch the code.

Reference · constraints.md (write it first)

```
# Refactor constraints
```

```
## May change
```

- Internal function structure, names, early returns.

```
## Must NOT change
```

- Public function signatures / CLI flags.
- File count (no new modules).
- Observable runtime behavior – tests must stay green.

```
## Style
```

- Replace nested conditionals with early returns.
- No comments unless they explain **why**.

Reference · Common mistakes

- Skipping `constraints.md` → Claude rewrites everything → lab spent reading.
- Combining refactor + docs in one prompt → docs describe the prompt.
- Vetoing every unrequested change (some are fine — read the diff).
- 200-line `ARCHITECTURE.md` (trim aggressively).

Live demo · Bad vs. constrained refactor (5 min)

1. Open `exercises/part-08/before/` (messy). Show the **unconstrained** refactor → bloated diff.
2. Reset. Paste the **constrained** prompt:

```
Refactor for readability only, obeying constraints.md exactly. Do not change public signatures, file count, or behavior. Tests must stay green. Show the diff.
```

3. Run tests → still green.
4. Pass 2: generate `HANDOFF.md` from the diff; read it aloud.

Success signal: the constrained diff respects every line of `constraints.md` and tests stay green.

▶ Your turn · Refactor + handoff docs (12 min)

Exercise: `exercises/part-08/README.md`

STEP 1 Copy the messy module to `module-08/after/`.

STEP 2 Write `constraints.md` **before** touching code.

STEP 3 Refactor for readability only; re-run the existing tests (must stay green).

STEP 4 Pass 2 — generate `HANDOFF.md` and `ARCHITECTURE.md` from the diff.

Success signal: tests green · diff respects every constraint · both docs within length limits.

✓ Done & next (1 min)

Definition of done

- ☐ [] Tests still green; diff respects every constraint.
- ☐ [] `HANDOFF.md` ≤ 40 lines, all four sections.
- ☐ [] `ARCHITECTURE.md` ≤ 80 lines, has a diagram + component paragraphs.

Next — we graduate from prompts to *agentic engineering*: Skills, Hooks, MCP, multi-agent.

Module 9 — Skills, Hooks, MCP & Multi-Agent Workflows.